

FFJM2 vs. BFGS on the MGH Problems

FFJM2 minimizes $\frac{1}{2}\|F(x)\|^2$ with a local quartic model of the residuals instead of a line search. At iteration k , the step d solves

$$M_k(d) = \mu\|d\|^2 + \frac{1}{2} \sum_i q_i(d)^2, \quad q_i(d) = r_i + J_i d + \frac{1}{2} d^\top H_i d,$$

where r_i and J_i are the i -th residual and its Jacobian row at x_k , H_i is the current approximation to the Hessian of that residual component, and $\mu\|d\|^2$ is a Levenberg-Marquardt penalty that damps the step in place of a line search. The candidate $x_k + d$ is accepted or rejected based on the ratio between the actual and the predicted reduction in f .

The first iteration starts at $\mu = 1$. Each accepted step divides μ by a fixed factor $\mu_{grow} = 10$, with a floor $\mu_{min} = 10^{-8}$, so μ keeps shrinking. Each rejected step instead multiplies μ by that same factor $\mu_{grow} = 10$ and re-solves the subproblem at the same point, until a step is accepted or μ reaches a ceiling $\mu_{max} = 10^{12}$. Unlike a scheme that resets μ at the start of every iteration, this value always carries over from the previous one. This carryover performed better in practice than restarting μ from scratch each time.

Every run reported below stops under one of three point-wise criteria, evaluated with the same default tolerances for `ffjm2` and for Optim.jl's BFGS [4] (kept equal so the comparison is fair):

- **Gradient:** the gradient norm reaches $\|\nabla f\| \leq 10^{-8}$. This is a genuine first-order stationary point.
- **Step:** the last step has (numerically) zero length. The solver stalls and stops making progress.
- **Function:** the objective value does not change between two consecutive iterations. The iterate is stuck on a plateau.

A run that exhausts the iteration budget without meeting any of the three would be reported separately as *not converged*, but no such case appears in the tables below; their “Criterion” column always reports one of the three above.

Table 1 compares the six `ffjm2` Hessian-update models, each run independently on the 34 MGH least-squares problems [5]. Its “Converged (gradient)” column counts how many of the 34 problems stopped under the gradient criterion; the remaining columns are mean $\pm$ standard deviation over the 34 problems. The models are plain BFGS and SR1, the Dennis-Wolkowicz update [1] (DW) and its interval-scaled variant [2] (IS+DW), interval-scaled BFGS [2] (IS+BFGS), and Powell-Symmetric-Broyden [6] (PSB).

Table 1: Comparison of the `ffjm2` Hessian-update models on the 34 MGH problems.

Model	Converged (gradient)	Iterations	Func. evals	Grad. evals	Time (s)
BFGS	29/34	56.88 $\pm$ 188.25	99.97 $\pm$ 351.85	57.88 $\pm$ 188.25	2.071 $\pm$ 6.121
SR1	32/34	23.12 $\pm$ 54.84	44.32 $\pm$ 112.92	24.12 $\pm$ 54.84	0.934 $\pm$ 2.992
IS+BFGS	29/34	44.00 $\pm$ 170.39	88.15 $\pm$ 343.34	45.00 $\pm$ 170.39	1.590 $\pm$ 5.763
DW	29/34	56.12 $\pm$ 181.91	103.24 $\pm$ 351.45	57.12 $\pm$ 181.91	1.865 $\pm$ 5.942
IS+DW	29/34	49.41 $\pm$ 173.04	93.74 $\pm$ 344.17	50.41 $\pm$ 173.04	1.768 $\pm$ 5.923
PSB	31/34	15.68 $\pm$ 22.53	31.38 $\pm$ 53.44	16.68 $\pm$ 22.53	0.424 $\pm$ 0.919

PSB and SR1 are the two best models: together they solve the most problems under the gradient criterion, and PSB in particular does so with by far the fewest iterations, evaluations, and execution time on average (Table 1). SR1 nonetheless certifies gradient convergence on one more problem than PSB (32/34 against 31/34). To see exactly where they disagree, Table 2 isolates the three problems on which at least one of the two methods does not stop under the gradient criterion.

Table 2: FFJM2 (PSB) vs. FFJM2 (SR1): per-problem comparison on the MGH problems.

Problem	Method	f	$\ \nabla f\ $	Func. evals	Grad. evals	Criterion
MGH10	FFJM2 (PSB)	4.40×10^1	5.75×10^{-5}	299	127	step
MGH10	FFJM2 (SR1)	4.40×10^1	5.50×10^{-2}	297	125	step
MGH16	FFJM2 (PSB)	4.29×10^4	9.26×10^{-5}	98	36	function
MGH16	FFJM2 (SR1)	4.29×10^4	6.26×10^{-5}	44	9	function
MGH31	FFJM2 (PSB)	1.53	2.53×10^{-8}	62	21	function
MGH31	FFJM2 (SR1)	1.53	4.39×10^{-9}	44	29	gradient

On MGH10 and MGH16, neither model reaches the gradient criterion: both stall under the step and function criteria, respectively, at essentially the same objective value, although FFJM2 with PSB reaches a smaller gradient norm on MGH10. MGH31 is the one case that favors SR1 over PSB. SR1 certifies a small gradient norm ($\|\nabla f\| \approx 4.4 \times 10^{-9}$), while PSB stops one iteration earlier under the function criterion, even though both reach practically the same objective value ($f \approx 1.5286$) and gradient norm. For this reason, PSB is the best choice between the two approaches when weighing cost against benefit.

FFJM2 (PSB) uses persistent decaying μ , a full Hessian reset on a bad ratio, and $-\nabla f + \text{Armijo}$ at $k = 0$ and as a safeguard. The following table compares it against BFGS with backtracking line search on the 34 MGH least-squares problems.

Table 3: FFJM2 (PSB) vs. BFGS (backtracking): per-problem comparison on the MGH problems.

Problem	Method	f	$\ \nabla f\ $	Func. evals	Grad. evals	Criterion
MGH01	FFJM2 (PSB)	2.38×10^{-22}	4.78×10^{-11}	11	8	gradient
MGH01	BFGS (backtracking)	1.98×10^{-22}	3.69×10^{-10}	53	40	gradient
MGH02	FFJM2 (PSB)	5.10×10^{-18}	3.84×10^{-9}	9	7	gradient
MGH02	BFGS (backtracking)	1.11×10^{-26}	2.20×10^{-12}	15	11	gradient
MGH03	FFJM2 (PSB)	3.29×10^{-11}	1.22×10^{-9}	33	21	gradient
MGH03	BFGS (backtracking)	5.05×10^{-30}	2.83×10^{-10}	233	174	gradient
MGH04	FFJM2 (PSB)	3.81×10^{-17}	8.73×10^{-9}	9	8	gradient
MGH04	BFGS (backtracking)	9.86×10^{-32}	4.44×10^{-10}	40	14	gradient
MGH05	FFJM2 (PSB)	1.31×10^{-16}	6.95×10^{-9}	9	8	gradient
MGH05	BFGS (backtracking)	2.50×10^{-22}	1.08×10^{-10}	22	19	gradient
MGH06	FFJM2 (PSB)	1.01×10^3	1.16×10^{-28}	10	2	gradient
MGH06	BFGS (backtracking)	1.01×10^3	1.16×10^{-28}	10	2	gradient
MGH07	FFJM2 (PSB)	1.49×10^{-17}	6.03×10^{-9}	16	10	gradient
MGH07	BFGS (backtracking)	5.24×10^{-19}	5.03×10^{-9}	41	30	gradient
MGH08	FFJM2 (PSB)	4.11×10^{-3}	1.65×10^{-9}	11	9	gradient

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Table 3: (continued)

Problem	Method	f	$\ \nabla f\ $	Func. evals	Grad. evals	Criterion
MGH08	BFGS (backtracking)	4.11×10^{-3}	7.39×10^{-10}	34	28	gradient
MGH09	FFJM2 (PSB)	5.64×10^{-9}	1.51×10^{-10}	8	7	gradient
MGH09	BFGS (backtracking)	5.64×10^{-9}	1.55×10^{-9}	6	5	gradient
MGH10	FFJM2 (PSB)	4.40×10^1	5.75×10^{-5}	299	127	step
MGH10	BFGS (backtracking)	4.40×10^1	2.28	552	386	step
MGH11	FFJM2 (PSB)	3.73×10^{-20}	5.34×10^{-10}	25	17	gradient
MGH11	BFGS (backtracking)	2.27×10^{-20}	7.17×10^{-10}	50	42	gradient
MGH12	FFJM2 (PSB)	4.56×10^{-18}	6.36×10^{-10}	13	11	gradient
MGH12	BFGS (backtracking)	1.58×10^{-21}	3.20×10^{-11}	53	41	gradient
MGH13	FFJM2 (PSB)	1.53×10^{-14}	6.45×10^{-9}	14	11	gradient
MGH13	BFGS (backtracking)	1.57×10^{-14}	3.61×10^{-9}	55	45	gradient
MGH14	FFJM2 (PSB)	9.42×10^{-19}	8.23×10^{-10}	10	7	gradient
MGH14	BFGS (backtracking)	3.52×10^{-22}	5.46×10^{-10}	44	31	gradient
MGH15	FFJM2 (PSB)	1.54×10^{-4}	9.06×10^{-9}	37	22	gradient
MGH15	BFGS (backtracking)	1.54×10^{-4}	4.13×10^{-10}	39	33	gradient
MGH16	FFJM2 (PSB)	4.29×10^4	9.26×10^{-5}	98	36	function
MGH16	BFGS (backtracking)	4.29×10^4	2.83×10^{-5}	72	26	function
MGH17	FFJM2 (PSB)	2.73×10^{-5}	2.07×10^{-10}	25	13	gradient
MGH17	BFGS (backtracking)	2.73×10^{-5}	8.23×10^{-9}	90	59	gradient
MGH18	FFJM2 (PSB)	1.85×10^{-17}	1.58×10^{-9}	79	45	gradient
MGH18	BFGS (backtracking)	2.83×10^{-3}	1.28×10^{-8}	44	41	gradient
MGH19	FFJM2 (PSB)	4.38×10^{-2}	1.42×10^{-9}	100	51	gradient
MGH19	BFGS (backtracking)	1.37×10^{-1}	8.29×10^{-10}	147	113	gradient
MGH20	FFJM2 (PSB)	1.14×10^{-3}	1.25×10^{-9}	12	10	gradient
MGH20	BFGS (backtracking)	1.14×10^{-3}	7.07×10^{-9}	49	39	gradient
MGH21	FFJM2 (PSB)	2.04×10^{-17}	7.09×10^{-9}	11	8	gradient
MGH21	BFGS (backtracking)	4.79×10^{-18}	1.50×10^{-9}	228	150	gradient
MGH22	FFJM2 (PSB)	1.54×10^{-14}	2.99×10^{-9}	13	10	gradient
MGH22	BFGS (backtracking)	1.65×10^{-13}	1.56×10^{-8}	136	112	gradient
MGH23	FFJM2 (PSB)	1.12×10^{-5}	1.08×10^{-9}	18	13	gradient
MGH23	BFGS (backtracking)	1.12×10^{-5}	1.21×10^{-8}	87	68	gradient
MGH24	FFJM2 (PSB)	1.73×10^{-6}	4.22×10^{-9}	36	24	gradient
MGH24	BFGS (backtracking)	1.73×10^{-6}	6.29×10^{-9}	588	450	gradient
MGH25	FFJM2 (PSB)	2.38×10^{-25}	1.36×10^{-11}	12	6	gradient
MGH25	BFGS (backtracking)	6.84×10^{-26}	7.27×10^{-12}	23	16	gradient
MGH26	FFJM2 (PSB)	1.40×10^{-5}	8.17×10^{-9}	28	16	gradient
MGH26	BFGS (backtracking)	1.40×10^{-5}	1.11×10^{-8}	36	34	gradient
MGH27	FFJM2 (PSB)	2.93×10^{-16}	3.87×10^{-9}	10	8	gradient
MGH27	BFGS (backtracking)	5.73×10^{-20}	3.69×10^{-9}	20	17	gradient
MGH28	FFJM2 (PSB)	2.20×10^{-15}	6.89×10^{-9}	9	8	gradient

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Table 3: (continued)

Problem	Method	f	$\ \nabla f\ $	Func. evals	Grad. evals	Criterion
MGH28	BFGS (backtracking)	2.53×10^{-16}	1.09×10^{-8}	26	18	gradient
MGH29	FFJM2 (PSB)	1.29×10^{-22}	2.04×10^{-11}	7	7	gradient
MGH29	BFGS (backtracking)	9.10×10^{-19}	1.56×10^{-9}	8	8	gradient
MGH30	FFJM2 (PSB)	7.33×10^{-21}	3.54×10^{-10}	9	7	gradient
MGH30	BFGS (backtracking)	1.97×10^{-19}	4.03×10^{-9}	46	23	gradient
MGH31	FFJM2 (PSB)	1.53	2.53×10^{-8}	62	21	function
MGH31	BFGS (backtracking)	1.34	4.48×10^{-9}	78	43	gradient
MGH32	FFJM2 (PSB)	5.00	3.85×10^{-16}	2	2	gradient
MGH32	BFGS (backtracking)	5.00	5.58×10^{-16}	2	2	gradient
MGH33	FFJM2 (PSB)	2.32	1.04×10^{-10}	14	5	gradient
MGH33	BFGS (backtracking)	2.32	1.42×10^{-10}	13	5	gradient
MGH34	FFJM2 (PSB)	3.07	1.55×10^{-10}	8	2	gradient
MGH34	BFGS (backtracking)	3.07	2.13×10^{-10}	9	2	gradient

The boxplot below compares the performance of the two approaches.

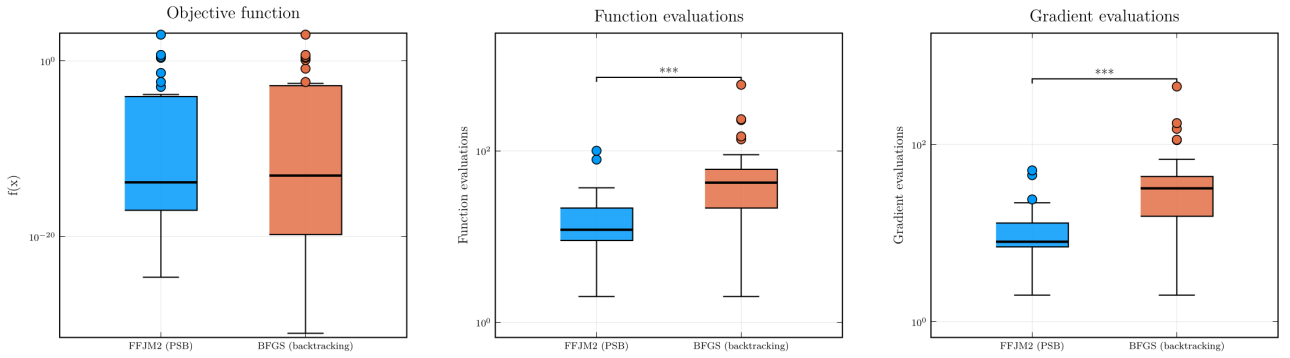


Figure 1: Comparison of FFJM2 (PSB) vs. BFGS (backtracking) on the MGH problems, restricted to runs where both methods converged by gradient norm. Final f value, function evaluations, and gradient evaluations. Stars indicate significance under the Mann-Whitney U test [3] ($p < 0.05$: *; $p < 0.01$: **; $p < 0.001$: ***).

As Figure 1 shows, the objective function values are not very different between the two methods: the p-value confirms neither achieves a significantly better result. Function evaluations and gradient evaluations, however, are different, and the p-value confirms this difference is significant. FFJM2 therefore performs better than BFGS in this respect, reaching a comparable solution at a much lower cost.

References

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